Structure formation in an expanding universe with a numerical-relativity particle-in-cell code: Working paper

Author list to be set
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This working paper records the extension of a numerical-relativity particle-in-cell code to cosmological structure formation. The aim is to redo, with that code, the comparison of general-relativistic and Newtonian collapse of East, Wojtak, and Pretorius, the integration tests of Daverio, Dirian, and Mitsou, and the initial-condition tests of Barrera-Hinojosa and Li. So far the code reproduces the homogeneous matter-dominated universe, the linear growth of a horizon-scale perturbation over two decades of expansion, and the luminosity distance along light rays in the homogeneous universe. Two findings concern the method. Matter sources held fixed during a spacetime step are a first-order error in an expanding universe. The deposit of a cold particle lattice carries an error set by its compression and by the smoothness of the particle cloud: proportional to the compression for the linear cloud of the method, to its square and cube for the quadratic and cubic clouds added here. In the collapse of a perturbation at one wavelength, run with identical particles and resolution in the two theories of gravity, the relativistic infall is later by a fraction of about three times the depth of the initial potential, a horizon-scale effect that is smaller than the delay reported by East, Wojtak, and Pretorius for the lower amplitudes, whose two codes differed in resolution. Light rays through the collapsing universe reproduce the luminosity distances of that work, and the integration tests of Daverio, Dirian, and Mitsou over a thousandfold expansion are passed for the homogeneous universe, with and without noise, and for a linear mode. The progress ledger in the appendix states what is done.

I. AIM

Cold dark matter is collisionless: after the first crossing of matter streams it has many velocities at one place, which a fluid cannot represent. A particle description coupled to the full Einstein equations follows it through the formation of bound structures, and answers how much a Newtonian N-body calculation on a prescribed expanding background misses. East, Wojtak, and Pretorius [1], building on Ref. [2], made that comparison for inhomogeneities at one wavelength with amplitudes from the cosmological to the extreme. Daverio, Dirian, and Mitsou [3] tested the time integration of a relativistic N-body code, and Barrera-Hinojosa and Li [4, 5] built and tested initial conditions for one.

The question of this work is whether the numerical-relativity particle-in-cell method of Ref. [6], as rebuilt in our earlier work for isolated systems, reproduces these results when it is turned to an expanding, periodic universe, and what the method needs for that. Section II writes the expanding universe in the variables of the code and shows why the calculation is inexpensive in conformal time. Section III gives the additions to the method. Section IV will hold the results, one subsection per source paper. Section V states two method developments that the first project called for.

II. AN EXPANDING UNIVERSE IN THE VARIABLES OF THE CODE

The code evolves the spatial metric γ_{ij} and the extrinsic curvature K_{ij} of the slices of constant time t, with lapse α and shift β^i, in the Z4c formulation, and moves macroparticles of rest mass m_p along geodesics [6]. A flat, matter-dominated homogeneous universe with scale factor a has

$$\gamma_{ij} = a^2 \delta_{ij}, \qquad K = -\frac{3}{\alpha}\frac{\dot{a}}{a} = -3H, \qquad H^2 = \frac{8\pi}{3}\rho, \tag{1}$$

where K is the trace of K_{ij}, a dot is a derivative with respect to t, H is the Hubble rate, and ρ is the density. Matter at rest in these comoving coordinates stays at rest, and its density falls as a^{-3}.

The slicing decides the cost. Gravitational fields travel at the speed of light, so a time step may cover at most a fraction of a cell. With the lapse condition

$$\partial_t \alpha = -\tfrac{1}{3}\,\alpha^2 K, \tag{2}$$

a member of the Bona–Massó family, the homogeneous solution has $\alpha = a$. The time coordinate is then conformal time, light covers one comoving cell per unit of $t/\Delta x$ at all epochs, and a fixed time step remains stable while the universe expands. In a matter-dominated universe $a \propto t^2$, so an expansion by a factor a takes $2(\sqrt{a}-1)/H_0$ of conformal time, where H_0 is the initial Hubble rate. For a periodic box of four initial Hubble radii, the choice of Ref. [1], the expansion to the collapse of a one-percent perturbation is a few box-crossing times, hence about a thousand time steps on a mesh of a hundred cells per side. The shift is zero. The mesh is uniform and periodic, and the spatial coordinates are comoving.

III. METHOD

A. Periodic domain and particles as observers

Particles that leave the box re-enter on the opposite side, and the clouds that deposit their energy, momentum, and stress on the mesh wrap around in the same way. Every particle carries its proper time τ, advanced with the lapse and Lorentz factor at the midpoint of each step, so that any particle serves as a fiducial observer. Particles of zero rest mass follow null geodesics with the same pusher and carry no sources; they serve as light rays. A ray that is emitted during the run waits at its point of emission, and at the emission time its direction becomes the null momentum of unit frequency for the normal observer there.

B. Matter sources in an expanding universe

The coupling of Ref. [6] deposits the energy density E, the momentum density S_i, and the stress S_{ij} seen by normal observers once per time step and holds them fixed during the stages of the spacetime integrator. Each is a sum over particles divided by the volume element $\sqrt{\gamma}$ of the cell. Around an isolated object the volume element changes little during a step. In an expanding universe it grows at three times the Hubble rate, and sources held fixed are a first-order error in time with that coefficient. The particle sums change only through the motion and the Lorentz factors of the particles, which is slow. The code therefore hands the spacetime solver the densities

$$\hat{E} = \sqrt{\gamma}\, E, \quad \hat{S}_i = \sqrt{\gamma}\, S_i, \quad \hat{S}_{ij} = \sqrt{\gamma}\, S_{ij}, \tag{3}$$

and the solver divides them by the volume element of each stage, $1/\sqrt{\gamma} = \chi^{3/2}$ with the conformal factor χ of the Z4c variables. The densities are extrapolated to the middle of the step from the last two deposits,

$$\hat{E}^{\,n+1/2} = \hat{E}^{\,n} + \frac{\Delta t_{n+1}}{2\,\Delta t_n}\left(\hat{E}^{\,n} - \hat{E}^{\,n-1}\right), \tag{4}$$

where Δt_n is the step that ends at the deposit n, and likewise for the other two, which makes the coupling second order for the slow part as well. Daverio, Dirian, and Mitsou refresh the same factor between particle updates [3].

The cloud that spreads a particle over the mesh smooths the deposited densities. A wave of wave number k on a mesh of spacing Δx is lowered by the factor $1 - \frac{1}{2}s^2(k\Delta x)^2$, where $s^2\Delta x^2$ is the second moment of the cloud in each direction: $s^2 = 1/6$, $1/4$, and $1/3$ for the linear, quadratic, and cubic cloud. The smoothing weakens the self-gravity of the wave and slows its growth. One pass over the deposited densities with the weights $(-c,\ 1{+}2c,\ -c)$ in each direction has the response $1 + 4c\sin^2(k\Delta x/2)$, and $c = s^2/2$ cancels the second moment, so that the density of a long wave is correct to fourth order in the cell size. The pass acts on the densities of Eq. (3), before the division by the volume element. The particles carry the density per coordinate volume, and on scales near the Hubble radius most of its contrast is the contrast of the volume element itself. The division is local, so a pass after it leaves that part of the smoothing in the sources. The pass also amplifies the sampling error of the deposit at the shortest waves, which limits its use once the flow is no longer smooth (Sec. **??**).

C. Time step and constraint damping

A fixed fraction of a cell is a stable step at all epochs, but its accuracy is set at the start of a run. The homogeneous solution changes on the expansion time $1/\mathcal{H}$, with $\mathcal{H} = \dot{a}/a$, and the integrator of the spacetime solver is of fourth order. Its error on the homogeneous background is committed in the first steps and stays in the solution as a uniform violation of the Hamiltonian constraint, which the expansion does not dilute. In a box several Hubble radii wide the Courant step is a sizeable fraction of the initial expansion time. The step is therefore limited to a fraction f of the expansion time,

$$\Delta t \le \frac{3f}{|\langle \alpha K \rangle|}, \tag{5}$$

with the mean over the mesh. The limit acts only while the expansion is fast, and the extra steps grow with the logarithm of the expansion over which it acts.

The constraint damping of the Z4c formulation acts at the rate $\kappa_1\alpha$ per unit of coordinate time. With $\alpha = a$ a constant coefficient κ_1 becomes stiff as the universe expands. The coefficient is divided by the lapse here, which damps the constraints at a fixed rate per unit of conformal time.

D. Initial data on the three-torus

The initial slice must satisfy the Hamiltonian and momentum constraints. Following Refs. [1, 2], the data start from a Newtonian density contrast δ_N and velocity v^i on a background with $a = 1$, with the potential ψ_N from $\partial^i\partial_i\psi_N = 4\pi\rho_0\delta_N$, and from the dictionary of Refs. [7, 8], which reads them as a metric and a matter distribution,

$$\begin{aligned} \mathrm{d}s^2 &= a^2\left[-(1+2\psi_N)\,\mathrm{d}t^2 + (1-2\psi_N)\,\delta_{ij}\,\mathrm{d}x^i\mathrm{d}x^j\right], \\ \frac{\rho}{\rho_0} &= 1 + \delta_N - 2\psi_N - 2\,\frac{\mathrm{d}\psi_N}{\mathrm{d}\ln a}, \\ K &= -3H\left(1 - \psi_N - \frac{1}{H}\frac{\mathrm{d}\psi_N}{\mathrm{d}t}\right). \end{aligned} \tag{6}$$

The source papers solve the conformal thin-sandwich equations for the corrections beyond linear order with the density held fixed. Here the particles are held fixed instead. The spatial metric is conformally flat, $\gamma_{ij} = \psi^4\delta_{ij}$,

and the extrinsic curvature is

$$K_{ij} = \psi^{-2}\hat{A}_{ij} + \tfrac{1}{3}\psi^4\delta_{ij}K,$$
$$\hat{A}_{ij} = \partial_i W_j + \partial_j W_i - \tfrac{2}{3}\delta_{ij}\partial_k W^k, \qquad (7)$$

with K from the dictionary and a vector potential W^i. The particle sums that the code deposits, $\hat{E} = \psi^6 E$ and $\hat{S}_i = \psi^6 S_i$ for the energy density E and momentum density S_i seen by normal observers, are fixed by the particle positions, masses, and momenta, up to the dependence of the Lorentz factor on ψ. The constraints become

$$\partial^j\partial_j\psi = \tfrac{1}{12}K^2\psi^5 - \tfrac{1}{8}\psi^{-7}\hat{A}_{ij}\hat{A}^{ij} - 2\pi\psi^{-1}\hat{E},$$
$$\partial^j\hat{A}_{ij} = \tfrac{2}{3}\psi^6\partial_i K + 8\pi\hat{S}_i. \qquad (8)$$

With the density scaling as the particles require, the linearized first equation has the operator $\partial^j\partial_j - \frac{9}{2}H^2$ on the homogeneous background, which is invertible on the torus; with the density held fixed the constant mode is marginal. The equations are solved on the mesh of the run with Fourier transforms, with the deposit of the code reproduced, so that the data satisfy the constraints for the sources the code will see.

The particles start on a lattice aligned with the mesh, with the lattice points at the cell corners, and are displaced by the Zel'dovich approximation exactly as in the Newtonian reference. The mass per coordinate volume must equal $\rho W\psi^6$ with Lorentz factor W, whose contrast is $\delta_N - 5\psi_N$ to linear order for growing-mode velocities; the particle masses carry the part of it that the displacement leaves out. For cold matter only the density and velocity fields enter the evolution, so the division of the mass among the particles is free. Identical positions give the two calculations the same sampling error of the deposit, which Sec. IV B shows to matter.

E. Newtonian reference

The Newtonian reference is a particle-mesh code on the mesh of the relativistic run with the same cloud shape, so that the two calculations differ in the gravity. Its output is turned into observer quantities with the dictionary above, as in Ref. [2], and light rays are integrated in the metric of the dictionary alongside the particles.

F. Observables

The comparisons use quantities defined by observers [1, 2]: the density $\rho(\tau)$ along the worldline of an observer comoving with the matter, expressed through

$$a_p = \left(1 + \tfrac{3}{2}H_0\tau\right)^{2/3}, \qquad \delta_{\rm obs} = \frac{\rho}{\rho_0}\,a_p^3 - 1, \qquad (9)$$

the proper time at which fiducial observers cross the overdensity, and the redshift and luminosity distance along light rays between the overdensity and the underdensity. For a ray with wave vector k^a, emitted and received by observers with four-velocity u^a that move with the matter,

$$1 + z = \frac{(u_a k^a)_{\rm emitted}}{(u_a k^a)_{\rm received}}. \qquad (10)$$

A bundle of rays that leaves the emitter within the solid angle $\mathrm{d}\Omega$ has the cross section $\mathrm{d}A$ where it is received, and the luminosity distance is

$$D_L = (1+z)\sqrt{\frac{\mathrm{d}A}{\mathrm{d}\Omega}}. \qquad (11)$$

Each bundle here is a central ray and four neighbors at a small opening angle; the cross section follows from the separations of the neighbors orthogonal to the central ray, and the solid angle from their directions in the frame of the emitter.

IV. RESULTS

A. Homogeneous expansion

A lattice of particles at rest in a periodic box of four Hubble radii must reproduce the homogeneous solution of Sec. II. The fields have no spatial dependence, so the test isolates the time integration of the coupled system. Figure 1 compares the two couplings of Sec. III B over a fourfold expansion. With the sources held fixed during a step, the lapse, the density, and the proper time of an observer drift from the exact solution at the percent level and the Hamiltonian constraint is violated at a tenth of its largest term. With the densities converted at each stage the errors fall by four to five orders of magnitude, to the level of the fourth-order time integrator, and the constraint stays satisfied.

B. Initial data and the sampling error of a lattice

The constraint solve of Sec. III D must return the dictionary to linear order. Figure 2 shows that it does so only for a suitable lattice. A particle at a cell center sits at the cusp of the cloud-in-cell weight. Where the displacement changes sign, at the density peak and at the trough, the two neighboring particles leak into each other's cells, the deposited contrast is half again too large there, and the conformal factor inherits the excess. With the lattice points at the cell corners, each particle is shared equally by eight cells, the deposit responds linearly to a small displacement, and the solved conformal factor agrees with the dictionary, with a remainder that is second order in the amplitude.

The same mechanism acts at any amplitude. A lattice whose spacing is compressed by the fraction c no longer has its harmonics on the zeros of the cloud shape, and

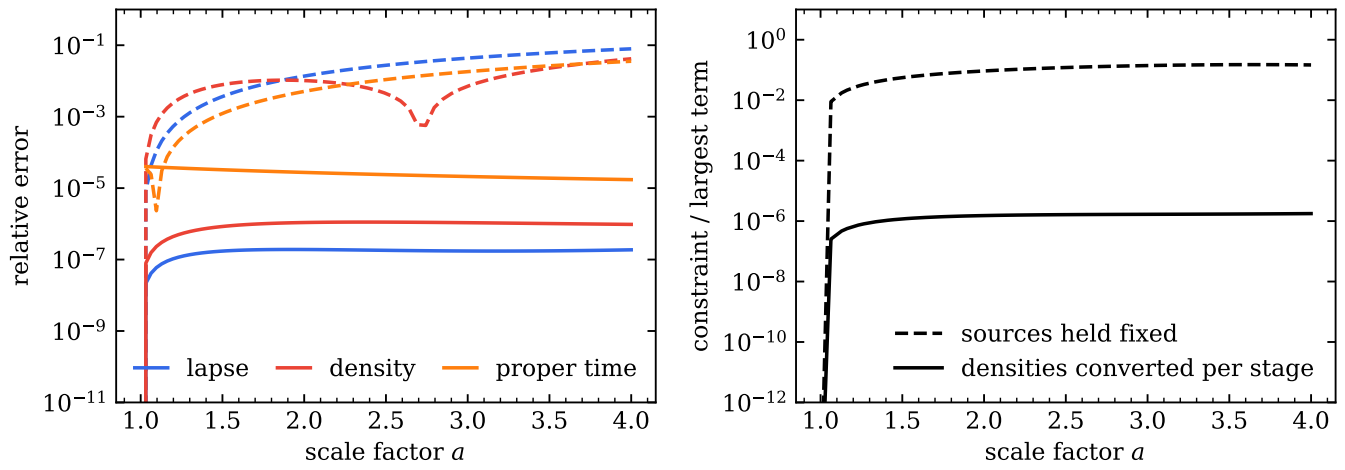


FIG. 1. Homogeneous universe. With the matter sources held fixed during a spacetime step (dashed), the lapse, the density, and the proper time of an observer drift from the exact solution; with the densities converted at each stage (solid) the errors are those of the time integrator. Left: relative errors against the exact matter-dominated solution; the proper volume of the box has the error of the density. Right: root mean square of the Hamiltonian constraint over the box in units of its largest term, $16\pi\rho$.

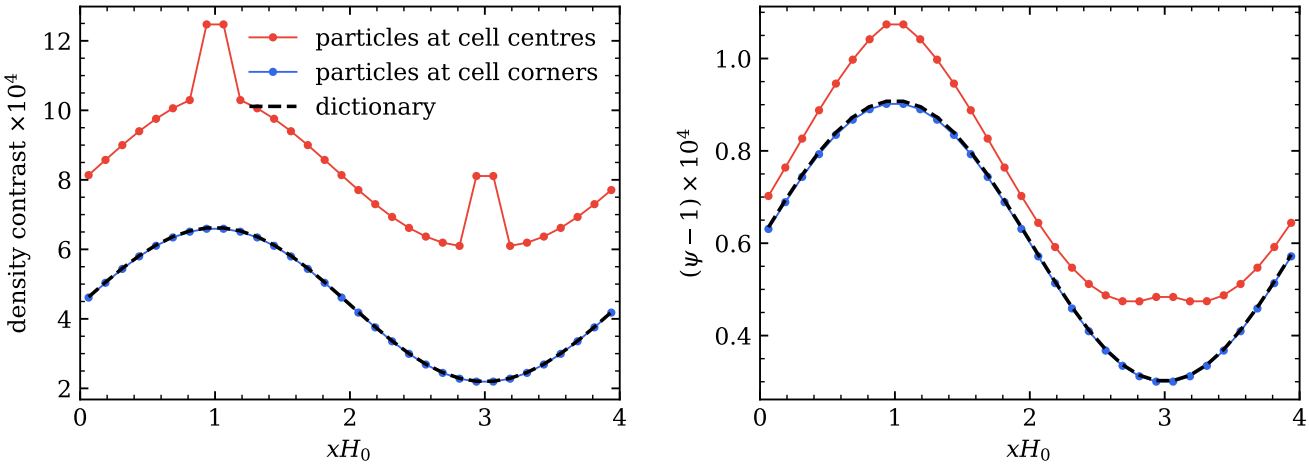


FIG. 2. Initial data at small amplitude along the mesh line next to the density peak. Particles at the cell centers (red) deposit a density contrast far above the dictionary value (dashed), with spikes where the displacement changes sign, and the solved conformal factor inherits the excess. Particles at the cell corners (blue) reproduce the dictionary. Left: deposited density contrast. Right: conformal factor.

each harmonic leaks into the deposit with the power of c set by the smoothness of the shape. On the mesh the leakage appears at the long wavelength h/c for cells of size h, which a filter of a few cells leaves in place, and it is largest where a particle meets a kink of the shape. Figure 3 shows the largest error of the deposited density for a uniform density sampled by a compressed lattice, for clouds that are B-splines of degree one to three. The error follows

$$\frac{c}{2}, \qquad 0.4\,c^2, \qquad 0.4\,c^3 \tag{12}$$

for the linear, quadratic, and cubic cloud with one particle per cell, and more particles per cell lower and cap it. With the linear cloud of Ref. [6] and one particle per cell, the density of a cold lattice is therefore wrong at the level of the compression itself, well before the collapse. A lattice point on a cell corner is a special case, with a deficit of c^2 per direction; the two density probes of the collapse problem sit on such points. A relativistic lattice displaced so that equal masses give the coordinate density is compressed four times more than the Newtonian one for the wavelength used here. This is the reason for the identical positions of Sec. III D, and the reason for the smoother clouds added to the code.

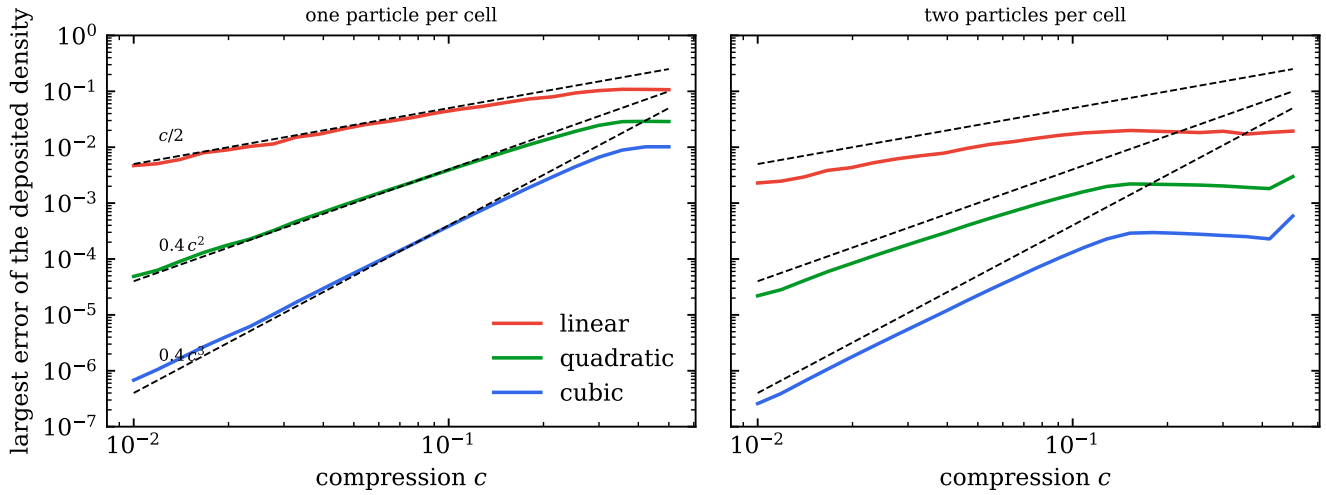


FIG. 3. Sampling error of a cold lattice. The largest error of the deposited density, for a uniform density sampled by a lattice compressed by the fraction c, falls with the smoothness of the particle cloud: it follows $c/2$, $0.4\,c^2$, and $0.4\,c^3$ (dashed) for the linear, quadratic, and cubic cloud with one particle per cell (left). Two particles per cell and direction lower and cap it (right). The deposit is filtered as in the runs.

C. Linear growth

For a small amplitude both calculations must follow linear theory. With the single-wavelength data of Ref. [1], $\delta_N = \sum_i \bar{\delta}_i \sin(kx^i)$ with $\sum_i \bar{\delta}_i = 3\bar{\delta}$, the growing mode has a constant potential, and the observer density contrast at the two extrema is

$$\delta_{\rm obs} = \pm 3\bar{\delta}\left[a_p + 3\left(\frac{H_0}{k}\right)^2 a_p^{-3/2}\right]. \qquad (13)$$

The first term is the growing mode. The second is what remains of the density offset $-2\psi_N$ of the dictionary once the slower clock of an observer inside the potential is taken into account; it decays. Figure 4 shows that the relativistic run, which starts from constraint-solved data and evolves in the slicing of Sec. II, and the Newtonian reference both follow this law over two decades of expansion, at both extrema, to within the size of the second-order terms. The lapse condition keeps the slices close to those of the Newtonian gauge in this regime: with zero shift it holds $\alpha\,\gamma^{-1/6}$ fixed in time, as the dictionary metric does for a constant potential.

D. Light rays in the homogeneous universe

In the homogeneous universe a ray emitted at the Hubble rate H_e has

$$D_L = \frac{2}{H_e}\,(1+z)^2\left(\sqrt{1+z}-1\right). \qquad (14)$$

Figure 5 compares bundles emitted at two times with this relation. The redshift agrees to the accuracy of the background. The luminosity distance is low by a small amount that falls along the ray, and that amount is the second-order time error of the pusher: the metric at the middle of a step is the mean of the two time levels, which lowers the coordinate speed of light by $(\mathcal{H}\Delta t)^2/8$ for the conformal Hubble rate $\mathcal{H}$ and time step Δt. The measured error coincides with this expression along every ray, so the launch, the cross section, and the solid angle of the bundles are right.

E. Collapse at one wavelength: general relativity against Newtonian gravity

With the two calculations started from identical particle positions on the same mesh, with the same cloud shape and time step, the comparison isolates gravity. Figure 6 shows the observer density contrast at the two extrema for the four amplitudes of Ref. [1], on a mesh of sixty-four cells per wavelength with one particle per cell. At the underdensity the two calculations coincide for the three lower amplitudes, as in Ref. [1], and separate only for the largest. At the overdensity the contrast rises steeply when the matter streams cross, and the rise comes later in the relativistic calculation, by an amount that grows with the amplitude.

The proper time at which a fiducial observer reaches the overdensity measures the delay without reference to coordinates [1]. Figure 7 follows observers that start at four distances from the overdensity. In both calculations they fall in, cross, and oscillate about the bound region, and the relativistic infall is later. Figure 8 shows the delay, the ratio of the proper-time scale factors at which the distance has fallen to a tenth of its initial value in the two calculations, against the amplitude. The delay is

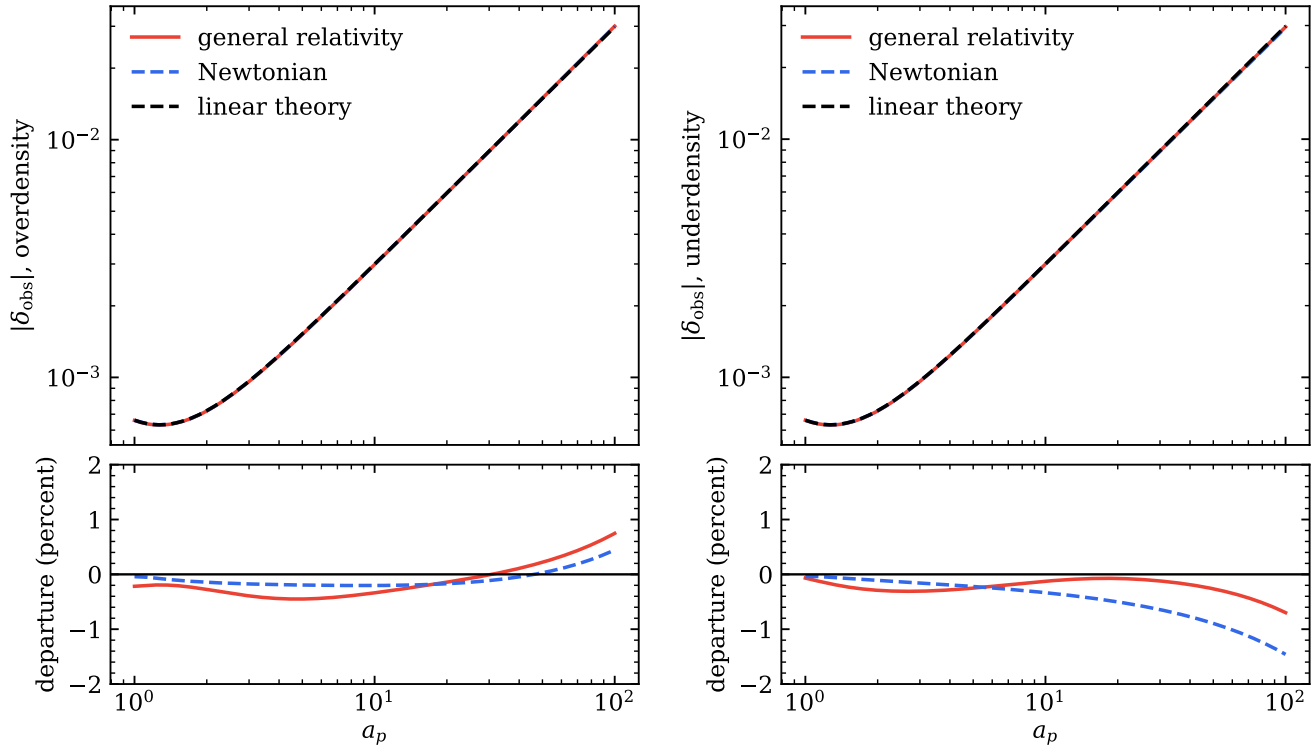


FIG. 4. Linear regime. The observer density contrast of the relativistic run (red) and of the Newtonian reference (blue, dashed) follows linear theory (black, dashed) over two decades of expansion at the overdensity (left) and the underdensity (right). Lower panels: departure from linear theory, which includes the smoothing of the density probe.

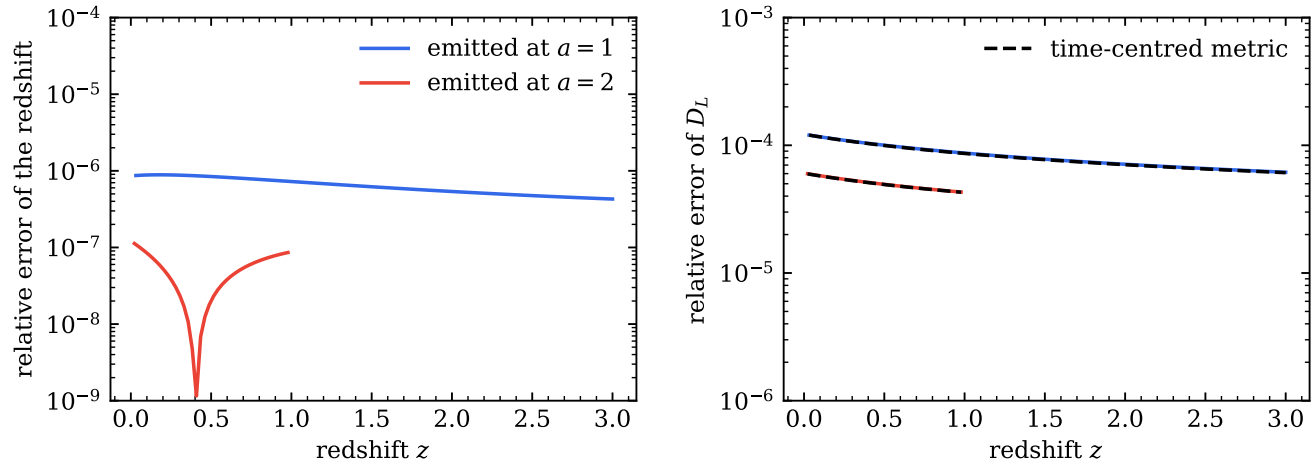


FIG. 5. Light rays in the homogeneous universe. The redshift along bundles emitted at two times agrees with the exact relation (left), and the error of the luminosity distance (right) coincides with the one expected of a pusher that centers the metric in time (dashed).

proportional to the amplitude over the whole range, the extreme case included. It is the same for the four starting distances and for the three cloud shapes of Sec. IV B. It is also the same when measured in coordinate time, and the clocks of the observers agree between the two calculations, so the infall itself is later. The dictionary that turns the Newtonian calculation into a spacetime is exact to first order in the perturbation, and the wavelength is several Hubble radii at the start; a difference of second order, proportional to the amplitude in relative terms, is what the comparison can show, and it shows one.

The delays read from the figures of Ref. [1] are larger

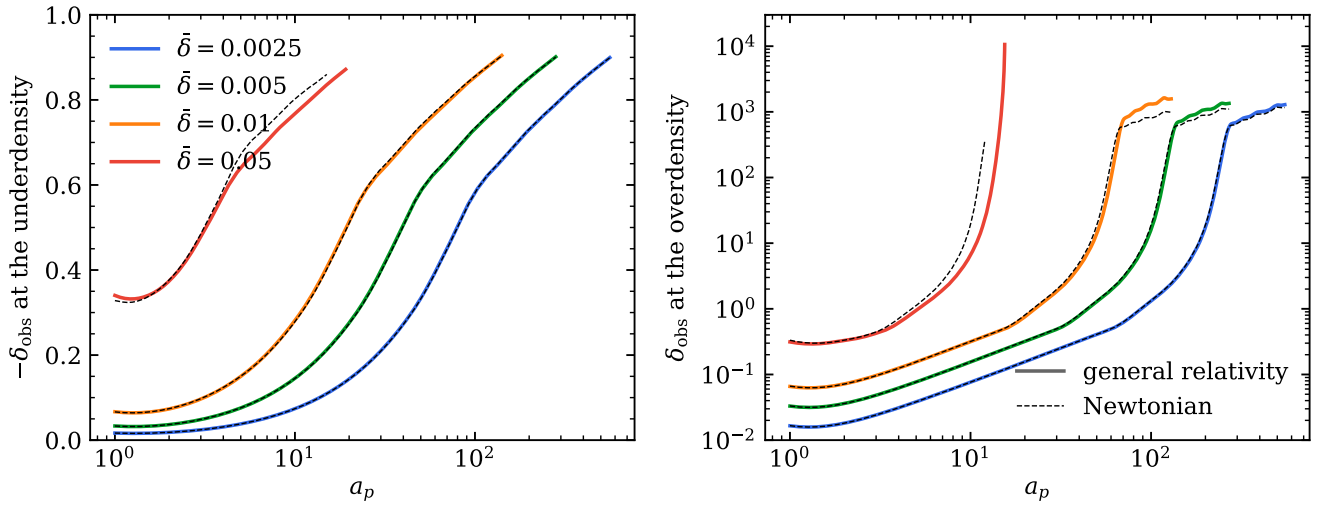


FIG. 6. Collapse at one wavelength. The observer density contrast of the relativistic run (colors) and of the Newtonian reference (black, dashed) coincides at the underdensity (left) for the three lower amplitudes; at the overdensity (right) the relativistic rise comes later, increasingly so for larger amplitude. The relativistic run at the largest amplitude ends when the lapse collapses at the overdensity, the Newtonian one where its potential reaches one half.

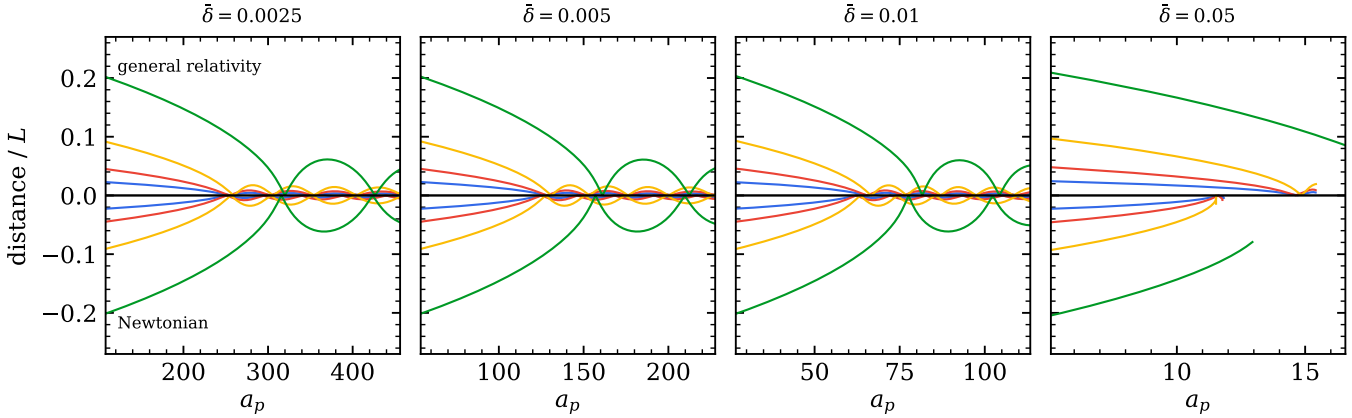


FIG. 7. Fiducial observers fall into the overdensity later in the relativistic run (upper half of each panel) than in the Newtonian reference (lower half, mirrored). Coordinate distance from the overdensity, in units of the wavelength L, of observers that start at $L/32$, $L/16$, $L/8$, and $L/4$ along one axis, against their own proper-time scale factor. At the largest amplitude both calculations end during the first infall.

for the three lower amplitudes and depend weakly on the amplitude. Its Newtonian reference had a finer force resolution than its relativistic code, and its text notes that finite resolution delays the collapse. With the same resolution in both calculations the delay is that of gravity alone for the given initial data.

The Newtonian calculation has no scale, so its infall does not depend on the wavelength; the relativistic one does. Repeating the comparison at one amplitude with the wavelength halved and doubled changes the delay by about a factor of four each way (Fig. 8, right): the delay follows the depth of the initial potential at the overdensity, about three times that depth, across all amplitudes and wavelengths run. It belongs to the horizon scale and vanishes for perturbations well inside the horizon, where the Newtonian calculation with the dictionary is accurate. At the longest wavelength the potential is as deep as for the extreme amplitude, and both calculations end near the collapse in the same way. How the delay divides between the second-order part of the initial data and that of the evolution is open: the nonlinear part of the constraint solve lowers the initial contrast at the overdensity by a fraction proportional to the potential, in the direction of a later collapse.

At the largest amplitude the lapse at the overdensity turns around and collapses while the density there rises without bound, and the run ends; the Newtonian dictionary fails slightly earlier, where its potential reaches one

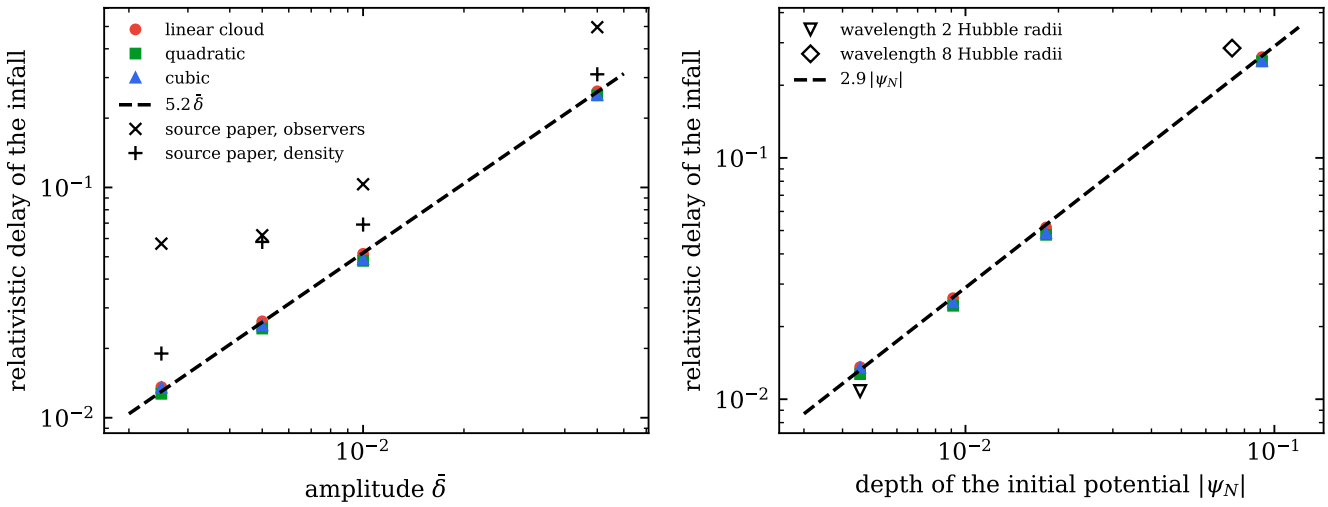


FIG. 8. The relativistic delay of the infall is set by the depth of the initial potential. Left: ratio, minus one, of the proper-time scale factors at which the distance of a fiducial observer has fallen to a tenth of its start in the relativistic and in the Newtonian calculation, averaged over the starting distances, against the amplitude, for the three cloud shapes; dashed: $5.2\,\bar{\delta}$; crosses: the same ratio and the ratio at a density contrast of ten, read from the figures of Ref. [1]. Right: the same delay against the depth of the initial potential at the overdensity, with two further wavelengths at the amplitude 0.01 (open symbols); dashed: $2.9\,|\psi_N|$.

half. Reference [1] reports the same end for both calculations and a black hole forming. The lapse condition of Sec. II holds $\alpha\,\gamma^{-1/6}$ fixed, so the lapse vanishes only with the volume element. A first alternative that holds $\alpha - \gamma^{1/6}$ fixed, and so stops the slices at a finite volume element, ends at the same place: the lapse then falls only as fast as the local scale factor. A driver toward constant mean curvature, which adds to the lapse equation a term proportional to the difference between the local K and its mean over the box, does what is needed: it vanishes in the homogeneous universe, leaves the observer quantities of the linear regime unchanged, and at the largest amplitude brings the lapse at the overdensity down to a few percent of its largest value while the density there grows slowly. That run ends later, when the collapsing region, a few cells across, is no longer resolved. With three levels of static refinement around the overdensity, onto which the initial data are interpolated, the collapse is followed to smaller scales and the lapse falls to a tenth of its largest value, but the spatial metric then degenerates in the innermost cells; a first shift driver, added against the stretching of the coordinates into the collapsing region, is itself unstable on the refined blocks. Following the black hole needs a shift condition that is stable there and initial data that satisfy the constraints on the refined blocks. This is the state of the extension.

F. Dependence on resolution

Reference [1] finds that the collapse comes earlier with resolution in both calculations and that the error at late times is set by the number of particles. Figure 9 repeats that study at one amplitude with the quadratic cloud and one particle per cell. The infall of the fiducial observers moves earlier by about a percent from sixty-four to ninety-six cells per wavelength and by a few parts in a thousand from there to one hundred and twenty-eight, in both calculations alike, and the relativistic delay changes by less than a fifth of a percentage point. The density reached in the bound region rises with resolution, as in Ref. [1]. The mean relative Hamiltonian constraint converges while the lattice is nearly undeformed and then becomes independent of the cell size: at one particle per cell the sampling error of Sec. IV B depends on the compression of the lattice and not on the mesh.

G. Constraint violation and the cloud shape

The collapse runs measure how the cloud shape acts on the constraints. Following Ref. [3], a constraint is reported relative to the root of the sum of the squares of its terms, H_r for the Hamiltonian and M_r for the momentum constraint. Figure 10 shows their means over the mesh. While the flow is single-valued the violation is set by the deposit. The deposited fields satisfy the continuity equations exactly as functions of position, and the finite differences of the spacetime solver see them through the mesh, so the mismatch is the part of the cloud that the mesh does not resolve. The quadratic cloud lowers the violation by one to two orders of magnitude and the cubic cloud by more, down to the level of the time integration at early times. Once the bound region has formed, the three shapes agree: the matter there has many velocities within a cell. The smoother clouds also remove a bias of

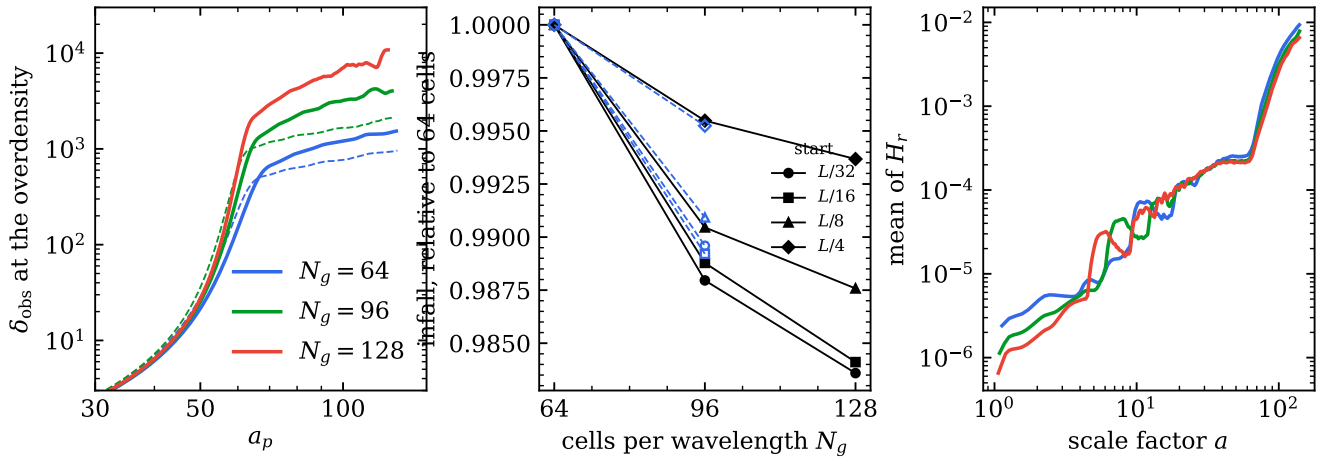


FIG. 9. Resolution study at the amplitude 0.01 with the quadratic cloud. Left: observer density contrast at the overdensity, relativistic (solid) and Newtonian (dashed); the density of the bound region rises with resolution. Center: scale factor of the infall of the fiducial observers relative to its value at 64 cells, relativistic (black) and Newtonian (blue, dashed), for the four starting distances. Right: the mean relative Hamiltonian constraint converges early and is set by the particle sampling later.

the collapse itself. The linear cloud holds the central density down, and the inner observers fall in several percent later than with the smoother clouds, in both calculations alike; the relativistic delay of Fig. 8 is unaffected.

H. Light rays through the collapsing universe

Light probes the spacetime along its path. Bundles of rays leave the overdensity toward the underdensity, and the reverse, at a set of emission times; the Newtonian bundles leave at the same proper times of the emitters and are integrated in the metric of the dictionary. Figure 11 shows the luminosity distance along each ray relative to the homogeneous relation of Sec. IV D for an emission at the proper time of the emitter. Rays emitted while the perturbation is linear deviate from that relation at the level of the perturbation, with opposite signs for the two directions. Rays that leave the overdensity after it has collapsed gain distance with little redshift at first, and rays from the underdensity are blueshifted as they fall into the bound region, where their redshift turns back. Both calculations show the same structure, and the deviations agree with those read from the figures of Ref. [1] to about a tenth of their size. The two theories of gravity differ far less than either differs from the homogeneous universe: by less than a part in a thousand for the rays emitted in the linear phase, and at the percent level only where a ray starts or ends in the bound region, the conclusion of Ref. [1].

I. Time integration over a thousandfold expansion

Daverio, Dirian, and Mitsou test the time integration of a relativistic particle code from redshift one thousand to today [3]. Their tests are repeated here with their box sizes, cell numbers, and noise amplitudes, in the units of this code.

In the first test the universe is homogeneous. With exact data the errors are those of the time integrator and converge at fourth order (Fig. 12, left). They are larger than in Ref. [3] and grow through the run. Part of this is the constraint variable Θ of the Z4c formulation, which enters the lapse condition and lets the lapse drift off conformal time while it decays; damping the constraints at a fixed rate per unit of conformal time, which amounts to dividing the damping coefficient by the lapse, removes that part. The rest is the truncation error of the first steps, which fixes the normalization of conformal time for the remainder of the run. With white noise on every field, the second test, the evolution is stable at all three resolutions and the largest constraint violations fall by seven orders of magnitude during the expansion, as in Ref. [3] (Fig. 12, center and right). The density error of the noisy runs grows more slowly than the scale factor, as required. Its level depends on the momentum noise of the particles: with the amplitude stated in Ref. [3] the particles drift freely by enough to exceed the density error reported there before any gravitational growth, and without momentum noise the early error equals the reported one. The runs of Ref. [3] are therefore matched by the second reading.

In the third test a plane wave of small amplitude starts outside the horizon and ends well inside it. The potential must stay constant and the density contrast must follow linear theory. Figure 13 shows both: the potential

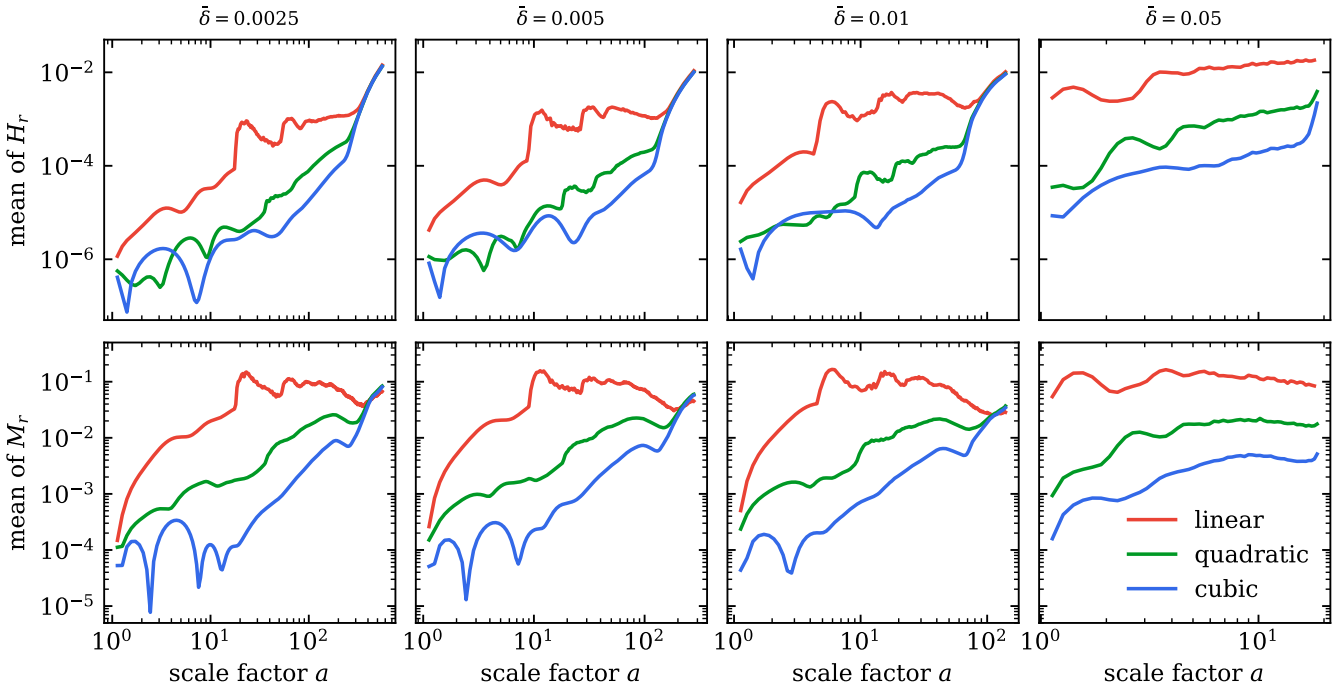


FIG. 10. A smoother particle cloud lowers the constraint violation of the collapse runs until the matter streams cross. Mean over the mesh of the relative Hamiltonian constraint (top) and momentum constraint (bottom) against the scale factor of the background, for the four amplitudes and the linear, quadratic, and cubic cloud.

changes by a fraction that falls by four per doubling of the cell number, and the departure of the density from linear theory converges at second order, below the values of Ref. [3] at equal cell number. The relative Hamiltonian constraint falls with resolution, where Ref. [3] reports it growing.

The fourth test, a random field drawn from a power spectrum, is being prepared.

J. Initial conditions with a spectrum of scales

Barrera-Hinojosa and Li derive the initial conditions of a relativistic particle code in a constant-mean-curvature slicing and test them at the initial time [5]. The particle positions must carry the particle-number contrast of the threading. On scales of the Hubble radius it exceeds the synchronous density contrast of a Newtonian simulation by a multiple of the curvature perturbation, which is constant in time. The velocities follow either from the growth rate or from the difference of two displacement fields at neighboring redshifts. All tests of Ref. [5] concern linear theory and the generator, so they are repeated here without evolution, for matter and a cosmological constant, with the same parameters, the same two boxes, and both starting redshifts. Every transfer function has a closed form in this case, and the generator is checked against the closed forms through the gauge transformations and the constraints.

Figure 14 shows three results. The growth rate f_p of the particle-number contrast falls from the matter value on small scales to zero beyond the Hubble radius. The closed form passes through the values of Ref. [5] up to the effect of radiation, which is absent here and lowers the growth rate of Ref. [5] on small scales by the factor of Mészáros. The power P_p of the particle-number contrast of the realized initial conditions, relative to the synchronous power P_S, equals the linear ratio on the realized modes in every bin, and agrees with Ref. [5] within the spread between realizations of the few largest modes. The finite-difference velocity carries more power $P_{\theta\theta}$ of the velocity divergence than the growth-rate velocity by the truncation error of the central difference, $2\,\Delta z^2/(1+z)^2$ for a redshift offset Δz, in every bin and in both gauges, as in Ref. [5].

Two statements of Ref. [5] are refined. The scatter it reports at the smallest redshift offset is reproduced by the rounding of tabulated input spectra and is far larger than single-precision arithmetic of the displacement field would give. For particles on a lattice the window correction of a mesh estimator is replaced by the linear response of a displaced lattice, which is derived and verified here.

The lapse condition of Sec. II keeps the Newtonian slicing while the potential is constant, which holds for the growing mode in matter domination, and zero shift keeps the threading in which the particle-number contrast is defined. The initial data of Sec. III D carry that contrast: the displacement supplies the synchronous part and the

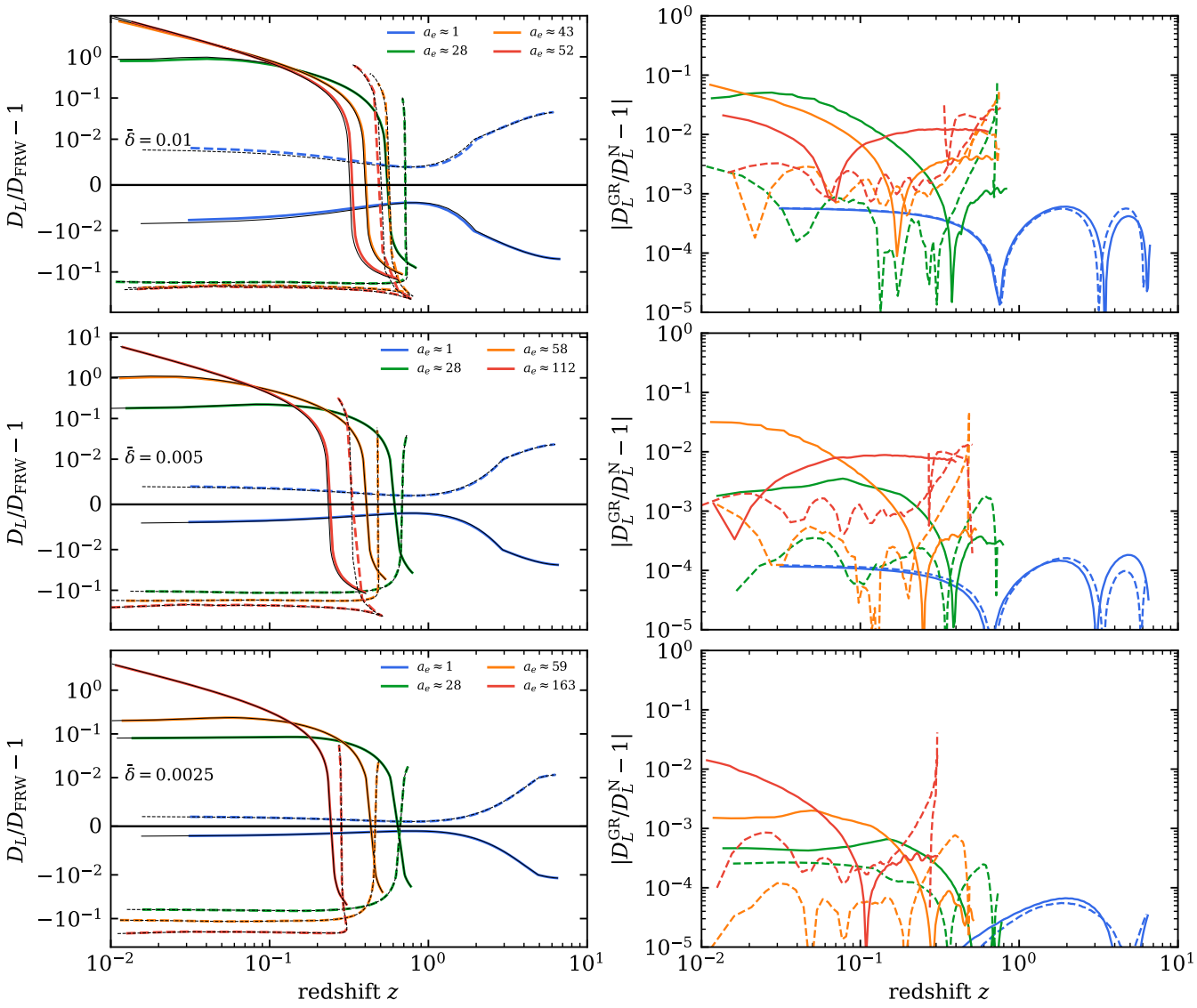


FIG. 11. Light rays between the overdensity and the underdensity deviate strongly from the homogeneous relation, and nearly equally in the two theories of gravity. Left: luminosity distance over the homogeneous relation, minus one, against the redshift, for rays emitted from the overdensity (solid) and from the underdensity (dashed) at the scale factors given; colors: relativistic run; thin black: Newtonian reference. Right: ratio of the relativistic to the Newtonian luminosity distance at equal redshift, minus one. The vertical axis on the left is linear between $\pm 10^{-2}$.

particle masses the remainder, and the constraint solve supplies the metric. They are therefore the counterpart of the initial conditions of Ref. [5] for a hyperbolic code. With a cosmological constant the slicing drifts from the Newtonian one late in the evolution, by the fraction of the potential by which the growth rate falls below one.

V. METHOD DEVELOPMENTS

The first project traced most of its difficulties to one coupling: the source on the mesh is only as smooth as the local cell size and particle number allow. Two developments address it. The first is a deposit whose momentum density satisfies, with the difference operators of the spacetime solver, the continuity equation of the deposited density, the gravitational counterpart of charge-conserving current deposition in plasma simulation. The second is to evolve with particles only the departure of the distribution from a known solution. In a cosmological run that starts from a lattice, the lattice already plays this role at early times. Both will be tested here once the baseline results exist. The cosmological runs already quantify the first: the sampling error of the deposit

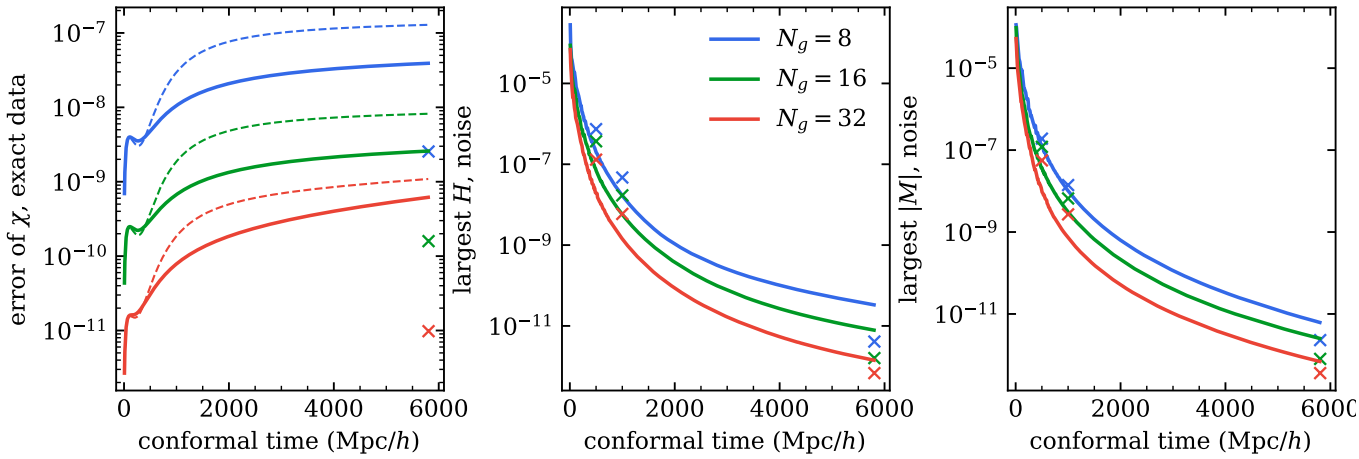


FIG. 12. Homogeneous universe from redshift one thousand to today at 8, 16, and 32 cells. Left: the error of the conformal factor for exact data converges at fourth order; dashed: constant damping coefficient; solid: damping at a fixed rate per unit of conformal time. Center and right: with white noise on every field the largest Hamiltonian and momentum constraint violations decay throughout the expansion. Crosses: values read from the figures of Ref. [3], in its units.

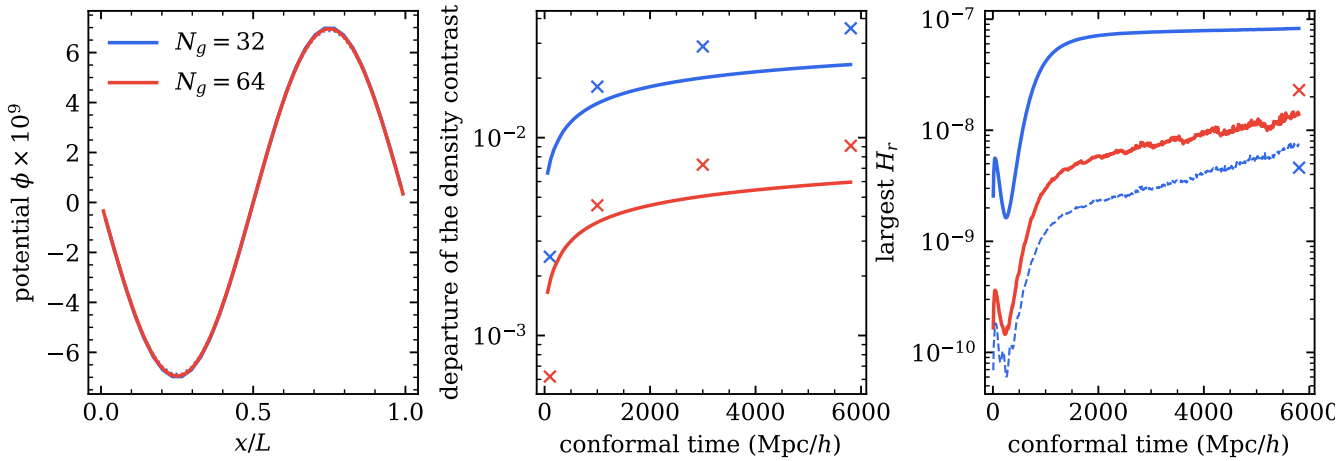


FIG. 13. Plane-wave linear mode from redshift one thousand to today. Left: the potential at the start (dotted) and at the end (solid) coincide. Center: the largest departure of the density contrast from linear theory, relative to the largest contrast, converges at second order. Right: the largest relative Hamiltonian constraint falls with resolution; dashed: 32 cells with the smaller time step of Ref. [3]. Crosses: values read from the figures of Ref. [3].

follows the compression laws of Sec. IV B, and the constraint violation of a run with smooth flow grows with it. The deposited fields satisfy the continuity equation exactly as functions of position; the finite differences of the spacetime solver see them through the mesh, and the mismatch is the part of the cloud shape that the mesh does not resolve. A smoother cloud is therefore the first step toward a constraint-compatible deposit, and its effect on the constraints is being measured.

Appendix A: Progress ledger

This appendix belongs to the working paper only. Table I lists the targets and their state.

Numbers behind the statements of the main text. C0, thirty-two cells, a quarter-cell step, fourfold expansion. Densities converted per stage: lapse within 2×10^{-7}, density and proper volume within 1×10^{-6} of the exact solution; root-mean-square Hamiltonian constraint 2×10^{-6} of its largest term. Sources held fixed: lapse 8 % low, density 4 % high, constraint 0.15 of its largest term.

C1, amplitude 10^{-4}, thirty-two cells, to $a = 100$. The Hamiltonian-constraint norm of the code at the start is

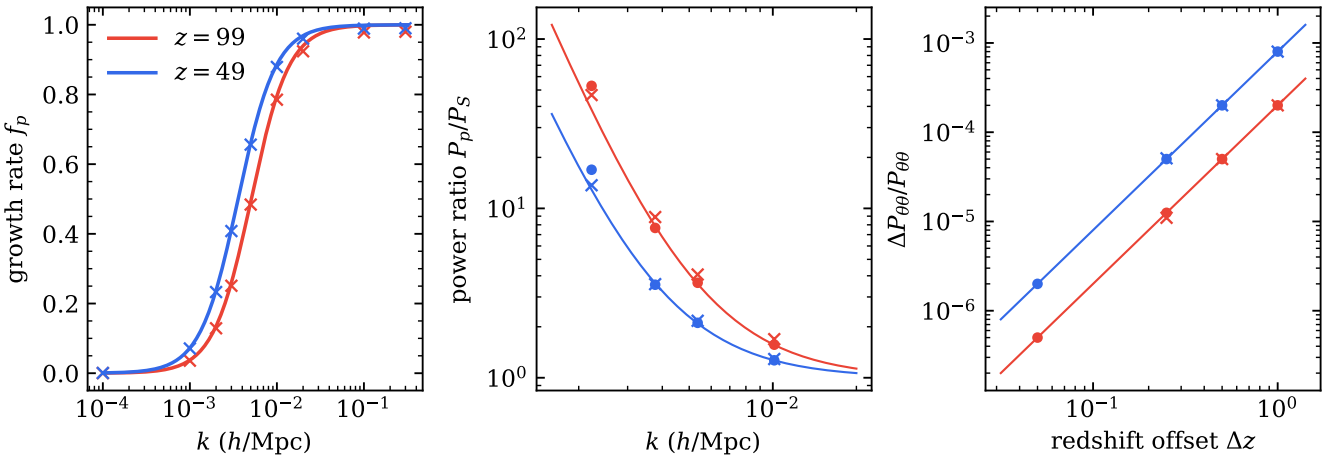


FIG. 14. Initial conditions with a spectrum of scales, at redshifts 99 and 49. Left: the growth rate f_p of the particle-number contrast vanishes beyond the Hubble radius. Center: the power of the particle-number contrast exceeds the synchronous power on the largest scales of a box of $4\,h^{-1}$Gpc; lines: linear theory at the bin center. Right: excess velocity power of the finite-difference velocity over the growth-rate velocity; lines: $2\,\Delta z^2/(1+z)^2$. Dots: this work. Crosses: values read from the figures of Ref. [5], which include radiation.

TABLE I. Targets and state. C0 to C6 are the milestones of the plan; T1 and T2 are the method developments of Sec. V.

Target	Content	State
Plan	Sources read; design of slicing, initial data, reference code, observables	done
C0	Periodic particles, cosmology setup, proper time per particle, homogeneous-universe test	done
C1	Constraint-solved initial data on the torus; linear growth	done
C2	Ref. [1]: observer density contrasts and crossing times, four amplitudes, both gravities	done at 64 cells, three cloud shapes
C3	Ref. [1]: luminosity distance against redshift	done at 64 cells
C4	Ref. [1]: constraint convergence, resolution and particle number, extreme case	cloud shapes, wavelength and resolution tests done (Newtonian run at 128 cells running); extreme case: lapse driver works; refined runs fail at the collapse
C5	Ref. [3]: all tests	homogeneous tests and linear mode done; random-field test: spectrum tools being written
C6	Ref. [5]: all tests	specification written
T1	Constraint-compatible deposit	first stage done: quadratic and cubic clouds, measured in the C2 runs
T2	Evolving the departure from a known solution	not started

6×10^{-10}. Conformal factor against the dictionary: 23 % of the perturbation off with particles at cell centers, 0.7 % with particles at cell corners. Observer density contrast: within 0.5 % of linear theory up to $a_p = 30$, +0.75 % and −0.70 % at $a_p = 100$ (contrast 0.03); the two theories differ by 0.3 % and 0.8 % of the contrast there.

C3 test, thirty-two cells. Redshift within 9×10^{-7}; luminosity distance low by 1.2×10^{-4} at emission at $a = 1$; measured and expected error agree to 6×10^{-7}.

C2, first runs (relativistic lattice displaced 4.04 times the Newtonian one). Deposited density at the probes 0.5 % below target at amplitude 0.01 and 7 to 14 % below at 0.05. First crossing of the fiducial observers later in the relativistic run by 5, 9, 14, and 23 to 28 % in a_p for amplitudes 0.0025, 0.005, 0.01, 0.05 (Ref. [1], read from its figures: 7, 9, 12 % for the three lower amplitudes). Largest amplitude: relativistic run ends at $a = 18.6$, Newtonian dictionary at $a = 14.4$. Constraint violation relative to its largest term up to $a = 4$: 3×10^{-4}, 1×10^{-2}, 1×10^{-1} for amplitudes 0.0025, 0.01, 0.05.

C2, identical positions, sixty-four cells, linear cloud. Contrast at $a_p = 1$, relativistic against Newtonian: −0.33, −0.53, −0.94, −5.8 % at the overdensity and +0.06, +0.24, +0.60, +3.7 % at the underdensity for am-

plitudes 0.0025, 0.005, 0.01, 0.05. Densities at the underdensity agree within 0.5, 1.0, 2.2 % for the three lower amplitudes. Delay of the infall: 1.37, 2.60, 5.15, 26 % ($5.2\,\bar{\delta}$); in coordinate time the same to 0.1 % of the scale factor. By cloud shape at amplitude 0.01, inner and outer observers: 5.2 and 5.3 (linear), 4.6 and 5.3 (quadratic), 4.7 and 5.3 % (cubic). Infall of the inner observers, relativistic run, amplitude 0.01: $a_p = 63.1$, 58.9, 59.6 for the three shapes. Read from the figures of Ref. [1]: 6 to 7, 6, 10 to 11 % for the observers and 1.9, 5.8, 6.9, 31 % at a density contrast of ten.

Constraint violation, amplitude 0.01, mean over the mesh up to $a = 4$: $H_r = 2.0\times10^{-4}$, 6.0×10^{-6}, 9.9×10^{-6} and $M_r = 3.8\times10^{-2}$, 1.6×10^{-3}, 1.9×10^{-4} for the linear, quadratic, cubic cloud; after the collapse $H_r \approx 10^{-2}$ for all.

Resolution, amplitude 0.01, quadratic cloud, 64, 96, 128 cells: relativistic infall of the inner observers at $a_p =$ 58.88, 58.18, 57.92 and of the outer ones at 79.36, 79.00, 78.86; Newtonian 56.29, 55.70 (inner) and 75.40, 75.04 (outer) at 64 and 96 cells; delay 4.61 and 4.44 % (inner), 5.25 and 5.27 % (outer). Mean H_r at $a = 2$, 10, 40: 4.7×10^{-6}, 6.4×10^{-5}, 2.3×10^{-4} at 64 cells and 1.8×10^{-6}, 4.5×10^{-5}, 2.1×10^{-4} at 128.

Wavelength test, amplitude 0.01, quadratic cloud: delay 1.1, 4.6 to 5.3, 28 to 30 % at 2, 4, 8 initial Hubble radii; delay over the depth of the initial potential 2.4, 2.75 (2.85 at amplitude 0.05), 3.9.

C3, sixty-four cells, linear cloud. Deviation of the luminosity distance from the homogeneous relation at $z = 0.1$, 0.5, 1, 2, relativistic run, with the values read from Ref. [1] in brackets. Amplitude 0.01, overdensity ray emitted at $a = 1$: −0.0069, −0.0043, −0.0042, −0.0108 (−0.0077, −0.0051, −0.0048, −0.0111); emitted at $a = 28$: +0.654, +0.025 (+0.605, +0.006); underdensity ray emitted at $a = 28$: −0.176, −0.189 (−0.161, −0.173). Amplitude 0.0025, overdensity ray emitted at $a = 163$ at $z = 0.1$: +0.320 (+0.309). Relativistic against Newtonian for rays emitted at $a = 1$: at most 6×10^{-4}, 2×10^{-4}, 7×10^{-5} for amplitudes 0.01, 0.005, 0.0025; largest for later rays at amplitude 0.01: 5 % (overdensity ray, $z = 0.03$) and 8 % (underdensity ray, $z = 0.73$).

C5. Homogeneous universe to $a = 1001$ at 8, 16, 32 cells, exact data: error of χ at $t = 100\,\mathrm{Mpc}/h$: 3.9×10^{-9}, 2.4×10^{-10}, 1.6×10^{-11} (fourth order); at the end 1.3×10^{-7}, 8.2×10^{-9}, 1.1×10^{-9} with the constant damping coefficient and 3.9×10^{-8}, 2.6×10^{-9}, 6.2×10^{-10} with damping at a fixed rate per conformal time (Ref. [3]: 2.6×10^{-9}, 1.6×10^{-10}, 9.8×10^{-12}). With noise: stable; largest Hamiltonian constraint from 1.2×10^{-5} at $t = 100$ to 3×10^{-11} at the end at 8 cells. Density error at the end at 8, 16, 32 cells: 6.9×10^{-4}, 2.3×10^{-4}, 6.5×10^{-5} with the stated momentum noise, 2.4×10^{-6}, 1.2×10^{-6}, 6.4×10^{-7} without it (Ref. [3]: 8.9×10^{-7}, 1.9×10^{-7}, 6.7×10^{-8}); at $t = 10$ without it: 9.0×10^{-9}, 2.3×10^{-9}, 5.5×10^{-10} (7.7×10^{-9}, 2.2×10^{-9}, 4.4×10^{-10}). Linear mode at 32 and 64 cells: potential constant to 1.7 and 0.44 % (4.9 and 1.0 %); density within 2.3×10^{-2} and 6.0×10^{-3} of linear theory at the end (3.6×10^{-2}, 9.1×10^{-3}); largest relative Hamiltonian constraint 8×10^{-8} and 1.4×10^{-8}.

Extreme case with the lapse driver (strength one), sixty-four cells: lapse at the overdensity down to 7.5 % of the largest lapse at $a = 20.5$; the run ends at $a = 20.6$ (without the driver: 18.6) when the metric determinant turns non-positive in 48 cells around the overdensity.

Appendix B: Stated differences from the source papers

The formulation is Z4c with the lapse condition above and zero shift; Ref. [1] uses the generalized harmonic formulation in a damped harmonic gauge. Comparisons therefore use observer quantities only. The initial data hold the particles fixed in the constraint solve, where Ref. [2] holds the density fixed; the total rest mass is that of the dictionary, and the density has the dictionary shape up to a constant factor that differs from one at second order in the amplitude. The Newtonian reference is a particle-mesh code at the resolution of the relativistic run, where Ref. [1] uses a tree code at higher force resolution. Particle numbers are lower than in the source papers, and the dependence on particle number is part of the tests.

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